GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form B · 29 points · two class periods

Name_____
Date_____
Period_____

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation

12 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(4.3 \times 10^{6}) + (2.4 \times 10^{6})$ *1 pt*

b. $(9.1 \times 10^{-5}) - (3.6 \times 10^{-5})$ *1 pt*

c. $(6.4 \times 10^{7}) + (8.2 \times 10^{7})$ *1 pt*

d. $(5.3 \times 10^{5}) + (2.6 \times 10^{4})$ *1 pt*

2. Multiply or divide. Write each answer in scientific notation.

a. $(4 \times 10^3)(2 \times 10^5)$ *1 pt*

b. $\frac{8.4 \times 10^8}{2.1 \times 10^3}$ *1 pt*

c. $(6 \times 10^{-4})(5 \times 10^6)$ *1 pt*

d. $\frac{1.5 \times 10^2}{5 \times 10^6}$ *1 pt*

3. Which is the value of $(2.3 \times 10^3) + (4.1 \times 10^5)$? *1 pt*

- ○ A. 6.4×10^5
- ○ B. 4.123×10^5
- ○ C. 4.33×10^5
- ○ D. 6.4×10^8

4. Select ALL of the expressions whose value is 4×10^6 . *1 pt*

- ☐ A. $(2.5 \times 10^6) + (1.5 \times 10^6)$
- ☐ B. $\frac{1.2 \times 10^{10}}{3 \times 10^4}$
- ☐ C. $(8 \times 10^3)(5 \times 10^2)$
- ☐ D. $(4.3 \times 10^6) - (3 \times 10^5)$
- ☐ E. $\frac{1.2 \times 10^{10}}{3 \times 10^3}$
- ☐ F. $(2.5 \times 10^6) + (1.5 \times 10^5)$

5. Devon writes $(3.4 \times 10^4) + (5.2 \times 10^6) = 8.6 \times 10^{10}$. Name the two mistakes he made, and give the correct sum. *1 pt*

6. A number is multiplied by 5×10^4 and the product is 3×10^{-2} . What was the number? Write your answer in scientific notation. *1 pt*

Part 2. Real-World Problems in Scientific Notation *4 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 3.5×10^2 mm by 1.25×10^2 mm. Find its area. *1 pt*

b. A vehicle travels 7.2×10^2 km in 8 hours. Find its average speed. *1 pt*

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 7.35×10^3 mm and 5.2×10^2 mm. Find their sum. *1 pt*

b. Two masses are measured: 8.4×10^{-2} g and 3.61×10^{-3} g. Find their difference. *1 pt*

Part 3. The Order of Operations with Exponents and Radicals *13 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $20 - 3 \cdot 2^2$ *1 pt*

b. $\frac{4^2 + 9}{5}$ *1 pt*

c. $45 \div 9 \cdot 3$ *1 pt*

d. $-3^2 - (-3)^2$ *1 pt*

10. Evaluate each expression.

a. $\sqrt{8^2 + 17}$ *1 pt*

b. $\sqrt[3]{-64} + 3^{-2}$ *1 pt*

c. $\left(-\frac{1}{2}\right)^2 + \sqrt{\frac{9}{16}}$ *1 pt*

11. A square patio of side 12 feet is sealed at 2.60 dollars a square foot, with a flat 75 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

b. Find the total cost. *1 pt*

12. Which is the value of $3 + 4 \cdot 3^2$? *1 pt*

- ○ A. 27
- ○ B. 63
- ○ C. 225
- ○ D. 39

13. Select ALL of the expressions whose value is 8. *1 pt*

- ☐ A. $\sqrt{64}$
- ☐ B. $64 \div 4 \cdot 2$
- ☐ C. $\left(\frac{1}{8}\right)^{-1}$
- ☐ D. $-\sqrt{64}$
- ☐ E. $\frac{5^2+7}{4}$
- ☐ F. $\sqrt{6^2+28}$

14. Tomas writes $\sqrt{36+64} = \sqrt{36} + \sqrt{64} = 14$. Explain why he is wrong, and give the correct value. *1 pt*

15. A square tile has an area of 225 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*